

AIStore Disclosure

This disclosure accompanies NVIDIA's MLPerf Storage submission and explains AIStore's project status, its relationship to existing storage products, and the intended interpretation of the submitted measurements.

Open-source NVIDIA technology

NVIDIA AIStore is not a managed service offering. It is open-source, MIT-licensed technology developed and used within NVIDIA. Its source code, documentation, deployment guidance, and deployment recipes are publicly available. Storage vendors, customers, researchers, and integrators are free to adopt, deploy, or integrate it under the license.

Flexible deployment with existing object storage

AIStore can operate as standalone persistent storage, as a high-performance fast tier, or as a unified namespace across multiple supported object-storage backends. A single AIStore cluster can connect to Amazon S3, Google Cloud Storage, Microsoft Azure Blob Storage, Oracle Cloud Infrastructure Object Storage, and remote AIStore clusters. In fast-tier deployments, AIStore brings frequently accessed data closer to compute while the existing backends remain the systems of record.

AI workloads

Training, inference, and data-processing clients



AIStore

Fast access through one unified namespace



Multiple backends

Amazon S3, GCS, Azure, OCI, and remote AIStore

This model is additive: adopting AIStore does not require replacing an organization's existing object-storage platform.

How to interpret the benchmark results

The submission reports **measurements from the specific deployments described in each System Description**, including their hardware, network, software versions, configuration, dataset, and test conditions. NVIDIA is publishing those measurements. This is not a comparative performance claim against third-party storage vendors. The results are not a product warranty, service-level commitment, or guarantee of performance in other environments; actual results may vary.

The submitted training portfolio includes AIStore deployments on **Oracle Cloud Infrastructure, Amazon Web Services, and Google Cloud Platform** through the S3-compatible API. These configurations are not like-for-like cloud comparisons.

REFERENCES

- [AIStore source code and MIT license](#)
- [AIStore architecture and fast-tier overview](#)
- [AIStore deployment guidance](#)
- [AIStore deployment recipes](#)
- [KubeCon 2024: AIStore as a Fast Tier Storage Solution for Petascale Deep Learning Across Cloud Backends](#)